

DANIEL BENJAMIN

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available starting May 2027 for a 4-month or 8-month co-op position

EDUCATION

University of British Columbia
Bachelor of Applied Science – Mechanical Engineering

Expected Graduation: May 2028

MECHANICAL DESIGN EXPERIENCE HIGHLIGHTS

- Designed and manufactured an [FSAE chassis frame](#) in SOLIDWORKS and developed Python scripts to calculate forces under various braking and cornering load cases for torsional rigidity simulations in ANSYS Mechanical
- Designed a [robotic hand](#) driven by soft pneumatic actuators 25% better at grasping than previous iterations
- Produced HVAC, plumbing, and fire suppression layouts in AutoCAD for commercial and residential projects
- Designed and validated multiple waterproof [3D-printed enclosures](#) for circuit boards against IPX5 standards

SELECTED EXPERIENCE

Chassis Team Lead (prev. Team Member), UBC Formula Electric Sep 2023 – Present

- Led 7-person chassis subteam using Agile project management with Jira and Gantt charts to coordinate deliverables and keep the 2026 competition vehicle on schedule
- Planned and executed carbon fiber + glass fiber composite layups for car body panels and aero kit
- Deployed SVN server on Oracle Cloud to track design changes alongside a [custom SOLIDWORKS integration](#)
- Designed a [custom 3D printer dashboard and Slack bot](#) to manage print jobs, status and reservations
- Wrote detailed design documentation and SOPs to ensure knowledge transfer for future team members

Mechanical Co-op, Avalon Mechanical May 2025 – Aug 2025

- Developed AutoLISP automation scripts to reducing CAD drafting time by over 30%, saving 7+ hours per week
- Assisted with drawing revisions, redlines, and documentation for tender and construction packages
- Performed hallway pressurization analysis and fan sizing using Excel and TRACE, calculating required airflow and pressure differentials to meet NFPA and provincial code requirements
- Validated fire sprinkler hydraulic systems using Elite Software FIRE for code-compliant designs

Undergraduate Research Assistant, UBC MEMS Lab May 2024 – Feb 2025

- Analyzed 15+ tendon-driven prosthetic hands to identify common successful design elements
- Rapidly prototyped design iterations using a custom-built multi-extrusion 3D printer running RepRapFirmware
- Designed a custom silicone return system around the actuators' force limits.
- Helped develop a demonstration platform for mimicking human hand gestures using MATLAB and Arduino

Volunteer, Victoria Hand Project July 2022 – Present

- Assembled and repaired 3D-printed low-cost prosthetic arms with voluntary open/close functionality
- Contributed concrete ideas to improve assembly speeds, leading to creation of new documentation

PROJECTS

WeBWorKer Chrome Extension github.com/danielrbenjamin/WeBWorKer

- Published an [extension](#) on the [Chrome Web Store](#) with over 900 users at peak adoption
- Used JavaScript and CSS to provide real-time LaTeX formatted previews of plaintext math on the WeBWorK homework platform, show matching parentheses, recognize variable names, and add custom query integrations

AI-Driven Predictive Maintenance System github.com/danielrbenjamin/manu465-project

- Applied [ML in Python](#) using TensorFlow to predict maintenance needs in smart manufacturing equipment

SKILLS

Mechanical: SOLIDWORKS, AutoCAD, ANSYS, Composites, Machining, 3D Printing, MIG Welding, FMEA

Software: C, MATLAB, Python, AutoLISP, JS, Arduino, Git, SVN, Docker, AWS, Jira, Bluebeam, Adobe CC